Local Silencing of Room Acoustic Noise using Broadband Active Noise Control

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Adaptive filtering techniques are now in widespread use for a number of applications such as adaptive arrays, adaptive noise cancellation, adaptive line enhancement, adaptive modeling and system identification, adaptive equalization, and adaptive echo cancellation [1]. More recently, these techniques have also been applied to the expanding field of active noise control [2]. In this paper, an application of active noise control to the silencing of acoustic noise in a room will be considered. First, a theoretical performance analysis using measured room impulse responses will be presented. Then an experiment will be described that uses a commercially available active noise control signal processor.

Figure 1 shows one possible configuration for an active noise control problem.

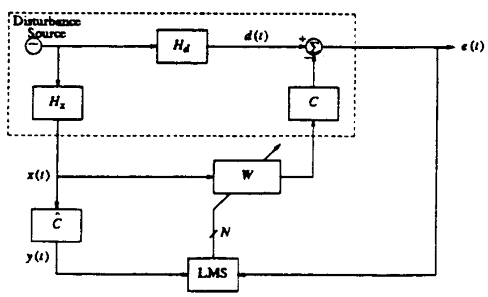

Figure 1. Generic block diagram of active noise control.

The system in the dashed box represents a plant over which we have no access except for the reference and error observation ports $x(t)$ and $e(t)$, respectively, and the control input which couples through the cancellation path transfer function C to cancel the disturbance signal $d(t)$. The disturbance and reference signals are assumed to be derived from a common disturbance source through linear transfer functions H_d and H_x, respectively. The intention is to adjust the coefficients of the FIR filter W in order to minimize the mean square residual error. Note that the filter W must have a sufficient number of taps to adequately span the combined length of H_d and $(H_xC)^{-1}$.

Adjustment of the adaptive filter coefficients utilizes the least-mean-square (LMS) algorithm [1], which is widely used in adaptive noise cancellation applications to remove undesired components of a primary signal that are correlated with a given reference signal. However, for active noise control applications, the canceling signal cannot be directly applied to the primary disturbance signal due to the intervening transfer function C, the nature of which can lead to instability [3]. One way to compensate for this effect is to filter the reference signal by an estimate of the cancellation path transfer function $\hat{C}$ before correlating with the error feedback signal, as suggested in [3]. This method was independently developed in [4], and is now commonly referred to as the "filtered-x" LMS (FXLMS) algorithm [1], which was independently conceived in the context of adaptive control systems [5]. The FXLMS algorithm and its IIR generalization have figured prominently in recent active noise control research [6]-[8].

As an example of the technique, consider the acoustic noise control problem of Fig. 2.

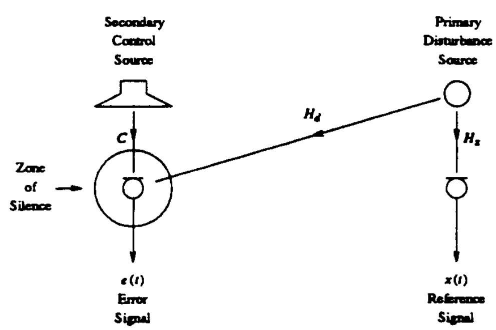

Figure 2. Schematic diagram of active noise control experiment.

The primary disturbance source on the right generates acoustic noise in an enclosed room. It is desired to create a "zone of silence" on the left side by sending a control signal to a loudspeaker which produces a canceling acoustic signal. A microphone on the left obtains the error signal $e(t)$ which is to be minimized. The reference signal $x(t)$ is derived from another microphone on the right, which is in close proximity to the primary disturbance. The various acoustic transfer functions are labeled in accordance with Fig. 1.

Impulse responses of the transfer functions in Fig. 2 were measured in an actual room (HuMaNet room A [9]) using SYSid [10] and truncated to 256 ms for the purpose of this theoretical analysis. For simplicity, a bandlimited white noise source was assumed over the band 100-500 Hz, which is sampled at 1 kHz. The theoretical minimum error was computed using the measured responses and is plotted in Fig. 3 as a function of the number of taps employed for the filter W.

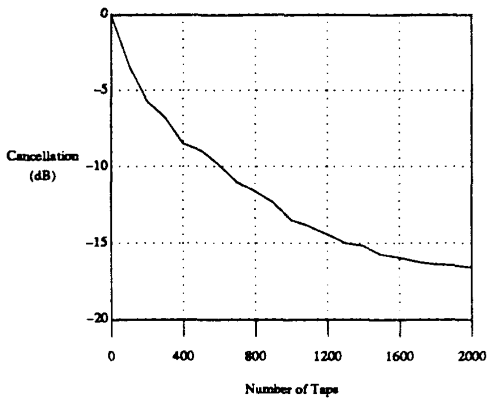

Figure 3. Theoretical active noise cancellation as a function of number of taps for 100-500 Hz, 10-th order bandpass noise.

As can be seen, several hundred taps are required to realize significant cancellation.

A wideband filter of 512 taps was selected as a compromise between performance and complexity, and a computer simulation of Fig. 2 was developed [11]. (The simulation assumes that acoustic feedback from the secondary loudspeaker to the reference microphone has been satisfactorily neutralized by other means.) Fig. 4 shows the resulting error spectra before and after convergence. As can be seen, more than 15 dB of cancellation is achieved over most of the band; the overall performance is dominated by the cancellation around 150 Hz and 400 Hz, the result being in rough agreement with the 9 dB level predicted from Fig. 3 for 512 taps.

Although good performance is expected at the error microphone, the utility of the technique

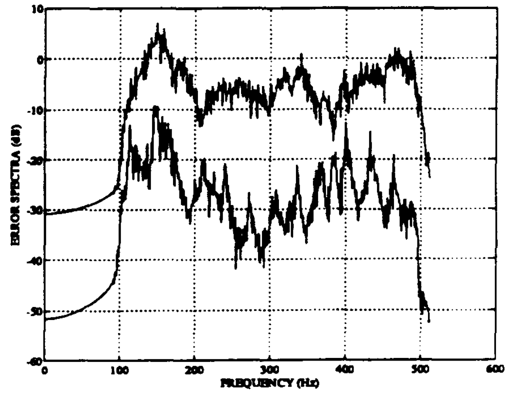

Figure 4. Error spectra of computer simulation before and after convergence.

depends on how large the zone of silence is. It is known that in a free-space noise field, good cancellation will be achieved only within a radius of a fraction of an acoustic wavelength [2]. Similar limitations apply in diffuse noise fields, where the proximity of primary and secondary sources is another factor [12]. As a rule of thumb, the radius of the 10 dB quieting zone is on the order of 1/10 of an acoustic wavelength. In air, the speed of sound is about 34 cm/ms. Thus, at 100 Hz, 10 dB of quieting might be expected within a 34 cm radius, while at 500 Hz, this characteristic distance reduces to 6.8 cm.

An experiment was designed to confirm the theoretical predictions of cancellation performance and to empirically determine the effective zone of silence that can be achieved. The setting of the experiment is in the same room that was used to measure the acoustic impulse responses used in the theoretical analysis and simulation. A Digisonix dX-57 Digital Sound and Vibration Controller is to be used to realize the adaptive filter shown in Fig. 1. This equipment additionally features on-line identification of the cancellation path transfer function C [13] that is incorporated as $\hat{C}$ in the reference path, as required for implementing the FXLMS algorithm; it also adaptively neutralizes the acoustic feedback from secondary source to reference microphone [7] (not shown in Figs. 1 and 2) which is an additional practical requirement. A white noise source with 100-500 Hz bandpass filter will be connected to a power amplifier feeding a loudspeaker on the right side of the room in order to simulate the primary disturbance source depicted in Fig. 2. A similar loudspeaker on the left side of the room will be used as a secondary source and omnidirectional microphones on the left and right side of the room will be employed for deriving the error and reference signals, respectively.

The goals of the experiment are to:

- Make measurements of the cancellation for a few different filter tap lengths to compare with the theory of Fig. 3.
- Make measurements of the error microphone spectra before and after convergence to compare with the computer simulation of Fig. 4.
- Make measurements of the residual acoustic spectrum using a third microphone at various distances from the error microphone in order to exhibit the effective zone of silence as a function of frequency.

The results of these experiments will be presented at the workshop.

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